SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

Code of Conduct

I N.Rama Sai Kiran Reddy [192525044] (Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks
Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Total Marks:

1) Define cloud-computing and explain four characteristics.

cloud computing is a technology that provides the computing resource as server, storage, databases, and software, and networking through internet.

Four characteristics

1) One-demand self-service: users can create or use cloud resource whenever without human assitance.

2) Broad network access: cloud services can be accessed through internet computers, tablets.

3) Resource pooling: The providers sharing computing resource multiple users according requirements.

4) Rapid elasticity: Resources can be increased or decreased based on demand.

Conclusion

These features make cloud computing more flexible, and scalable traditional on-prime computing.

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2) Trace the evolution of cloud computing

The evolution of cloud computing is happend through several stages:

1) Mainframe computing:

In the early days, many users shared a powerful central computer through terminals.

2) personal computers

computing became decentralized, with individuals using own computers.

3) Internet development

The growth of internet allowed computers and services communicate globally.

4) virtualization: physical servers could divided into multiple machines, improving utilization

5) web-based software:

Applications started being provided through web browers instead of requiring local instalation.

6) cloud computing

companies began providing computing resources over internet pay-as-you basis.

Technologies such as virtualization, high-speed internet, distribution computing, and data centers enabled modern cloud Aws, Microsoft.

Explain the role of server virtualization in cloud computing.

→ server virtualization is the process of diving one physical server into multiple servers software called hypervisor.

Role in cloud computing

→ Improves utilization to physical servers.

→ Allows multiple users to share the same hardware.

→ Reduces hardware and operating costs.

→ Makes it earier to create, move and manage virtual machines.

→ provides flexibility.

Virtualization	cloud computing
→ It is technology.	→ It service delivery model.
→ creates virtual resource from physical resource.	→ provides computing resource through the internet
→ can work without internet	→ usually accessed through a network.
→ Examples: VMware, VirtualBox.	→ Examples: AWS, Azure, Google cloud.

4) Discuss the role of data centers in delivery the cloud services.

A data centers is facility containing servers, storage systems, networking equipment, power system and cooling system.

Data centers play an important role in cloud computing because they:

→ store and process user data
→ Run cloude applications and virtual machines
→ Provide computing and storage resource.
→ Maintaining high avaliability of cloud services.
→ provides backup and disaster recovery
→ use security systems to protect data.
→ support scalability by adding more computing resource.

Cloud providers operate large data centers in the different locations so that can access services reliability quickly.

Differentiate Iaas, paas, saas, and xaas with examples.

Model	Meaning	what it provide	Example.
Iaas	infrastructure as service.	Virtual Machines	Amazon EC2.
paas	platform as a service	platform and tools.	Google App Engine.
saas	software as a service	Ready-to-use software	Google Docs.
Xaas.	Everything as a service.	Delivered through cloud	Database as a service.

Iaas

provides basic infrastructure. The user manages operating system and applications.

paas

provides a development platform. Developers can create can deploy applications underlying hardware.

saas

provides complete software users can directly through the internet.

xaas

A general term delivering and IT capability as a service.

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